

5 Practical Tips to Improve your Linux Log File Analysis Efficiently

18th July 2026



Grafana Labs



AlmaLinux

Agenda

1 Enabling & Configuring Logs Correctly

Fix the input before you analyze it, permissions & default configs

2 Command-Line Tools

CLI tools to query, don't scroll

3 Rotation, Retention & the TRACE Trap

Disk fires, and the debug level nobody remembers to turn off

4 Log Aggregation Tool

Centralized place to search logs across every host

5 From Logs to Alerts

Let the system tell you first, before the customer

6 Q&A



Presenter



Syed Usman Ahmad

Staff Developer Advocate



> \$ history | less

What I do?

My previous roles?

What I do beside work?



youtube.com/@freelinuxtutorials

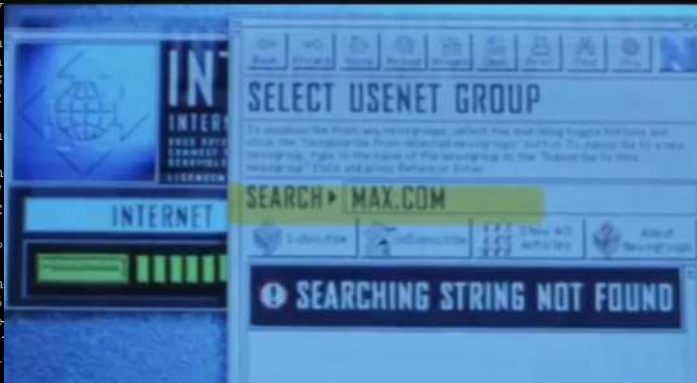


linkedin.com/in/usmanlinux/



Let's Talk about Logs

```
dEvtMsg", "meetingId": "fc0ee89b1e0d0ed83730414a0f7999b55", "userId": "w_w0lyusdelrfz"}, "body": {"intId": "w_w0lyusde assignedBy": "w_w0lyusdelrfz"}} emitted
t20181123-17:11:12.909(0) info: REDIS: PresenterUnassi
ync
t20181123-17:11:12.909(0) info: REDIS: {"header": {"nam
vtMsg", "meetingId": "fc0ee89b1e0d0ed83730414a0f7999b557f", "userId": "w_w0lyusdelrfz"}, "body": {"presenterId": "w_w0ly
me": "Bobak", "assignedBy": "w_w0lyusdelrfz"}} emitted
t20181123-17:11:12.909(0) info: Set a new presenter in
TATION POD meeting=fc0ee89b1e0d0ed83730414a0f7999b557fa
t20181123-17:11:12.909(0) info: Changed user role=PRES
z meeting=fc0ee89b1e0d0ed83730414a0f7999b557fa0292-1542
w0lyusdelrfz
t20181123-17:11:12.910(0) info: REDIS: PresenterAssign
c
t20181123-17:11:12.910(0) info: REDIS: {"header": {"nam
espMsg", "meetingId": "fc0ee89b1e0d0ed83730414a0f7999b557", "userId": "w_w0lyusdelrfz"}, "body": {"podId": "DEFAULT_PRE
esenterId": "w_w0lyusdelrfz"}} emitted
t20181123-17:11:12.911(0) info: REDIS: SetPresenterInP
nc
t20181123-17:11:12.911(0) info: REDIS: {"header": {"nam
dEvtMsg", "meetingId": "fc0ee89b1e0d0ed83730414a0f7999b55", "userId": "w_w0lyusdelrfz"}, "body": {"intId": "w_w0lyusde
assignedBy": "w_w0lyusdelrfz"}} emitted
t20181123-17:11:12.912(0) info: REDIS: PresenterUnassi
ync
```



Send the Logs

la.webloca.net

Logs in General

- 1 A digital footprint of user or system actions
- 2 Tracking with a timestamp
- 3 Trustworthy source for auditing

**“A watchtower is pointless if there’s
no watchman inside”**

The MANY Types of Logs

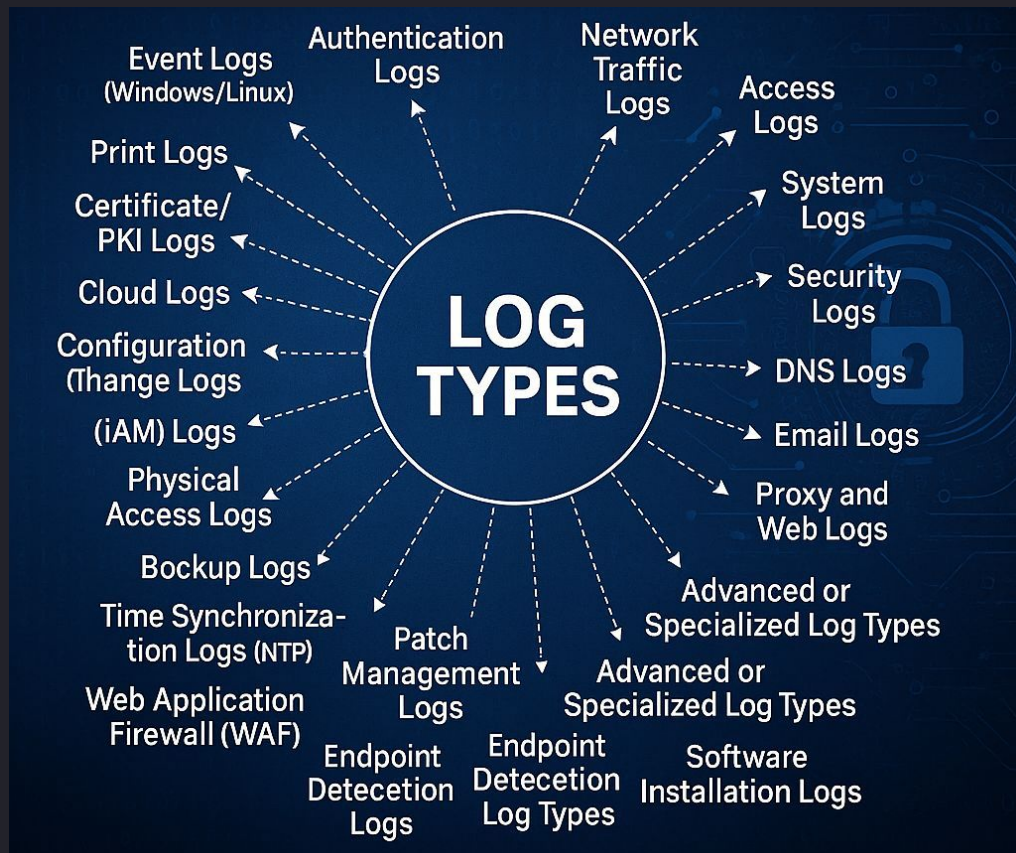


Image source:
<https://shorturl.at/srlk2>



Challenges

- 1 Not setup correctly! (*making basic mistakes*)
- 2 Dependency on external tools (*expensive*)
- 3 Setup & Config, no one remembers!
- 4 Lack of centralized visibility, accessibility

“and much more”

5 Practical Tips to Improve Your Linux Log File Analysis Efficiently

Tip #1
Enabling &
Configuring
Logs Correctly

Two Mistakes That Poison Everything Downstream

MISTAKE #1

Incorrect Permissions

The log destination is owned by the wrong user. The app fails to write, happens silently and nobody notices until the log they need simply isn't there.

MISTAKE #2

Trusting the Defaults

Most services logs are far less than you think out of the box. The detail you need during an incident was never being captured in the first place.

Example: Incorrect Permissions on the Log Destination

- App runs as **appuser**, but the log directory is owned by **root:root** at **640** (every write fails).
- Many apps swallow the write error silently.

WHY IT MATTERS

This is the #1 root cause of "we have no logs from right when it broke." Perform a permissions check into deployment.

```
$ ls -l /var/log/myapp/
-rw-r----- 1 root root 0 app.log
$ sudo -u appuser sh -c 'echo hi >> app.log'
sh: app.log: Permission denied
# fix:
$ chown appuser:appuser app.log
$ chmod 640 app.log
```

Example #2: Trusting the Defaults (MySQL)

- Out of the box, MySQL enables only the error log (mysqld.log).
- General query log and slow query log are **OFF** by default, therefore the exact detail you need mid-incident isn't being captured.

THE HABIT TO BUILD

For every service you run (database, web server, app framework) read the logging section of the docs, before you need it during an outage.

```
$ mysql -e "SHOW VARIABLES LIKE '%log%';"
```

```
general_log      OFF
```

```
slow_query_log   OFF
```

-- enable before you need it:

```
mysql> SET GLOBAL general_log = 'ON';
```

```
mysql> SET GLOBAL slow_query_log = 'ON';
```

```
mysql> SET GLOBAL long_query_time = 1;
```


Tip #2
Using
Command Line
Tools (CLI)

Using Command Line Tools (CLI)

- 1 Core CLI -> first point of contact
(*no external dependency!*)
- 2 Helps to track patterns, symptoms, anomalies
- 3 Example: less, tail, etc. in combination with grep, awk, sed etc. (also redirecting)

Example: Using core tools tail, grep, awk, sed

- No install, no dependency. Available on every Linux box you'll ever SSH into.
- Your first point of contact: fast enough to spot a pattern, symptom, or anomaly before reaching for anything heavier.
- The real power is in the piping: Chaining small tools into exactly the view you need.

```
$ tail -f /var/log/myapp/app.log           # live view
```

```
$ grep -ri "error" app.log | tail -20      # last 20 errors
```

```
$ awk '{print $9}' access.log | sort | uniq -c | sort -rn | head -5
```

```
42 500
```

```
9 502
```

```
3 504
```

```
# viewing HTTP status codes
```

```
$ sed -n '/09:15:00/,/09:20:00/p' app.log > incident.log
```

```
# isolate the incident window
```


Tip #3
The RRL Trap

Rotation, Retention & Level Trap (RRL)

- 1 The most **innocent**, but the most **costly**!
- 2 You got an escalation from a customer, but you don't have logs during that timestamp
- 3 SLA breach during **Sev 1** and **Sev 2**

Example #1: When Nobody Configures Rotation

- A verbose service can write Gigabytes of logs in few hours.
- Disk hits 100% → the logging service crashes, then dependents crash right after.
- The classic 3 AM pager about "disk full". Why → is really "log rotation was never configured."

```
$ df -h /var/log
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sda1	20G	20G	0	100%	/var/log

```
$ du -sh /var/log/myapp/*.log
```

```
14G /var/log/myapp/app.log
```

```
$ ls /etc/logrotate.d/myapp
```

```
ls: cannot access '/etc/logrotate.d/myapp': No such file or directory
```

The Fix: A Real logrotate Policy

```
/etc/logrotate.d/myapp
```

```
{  
    daily  
    rotate 14  
    size 500M  
    compress  
    delaycompress  
    missingok  
    notifempty  
    copytruncate  
}
```

WHAT EACH LINE BUYS YOU

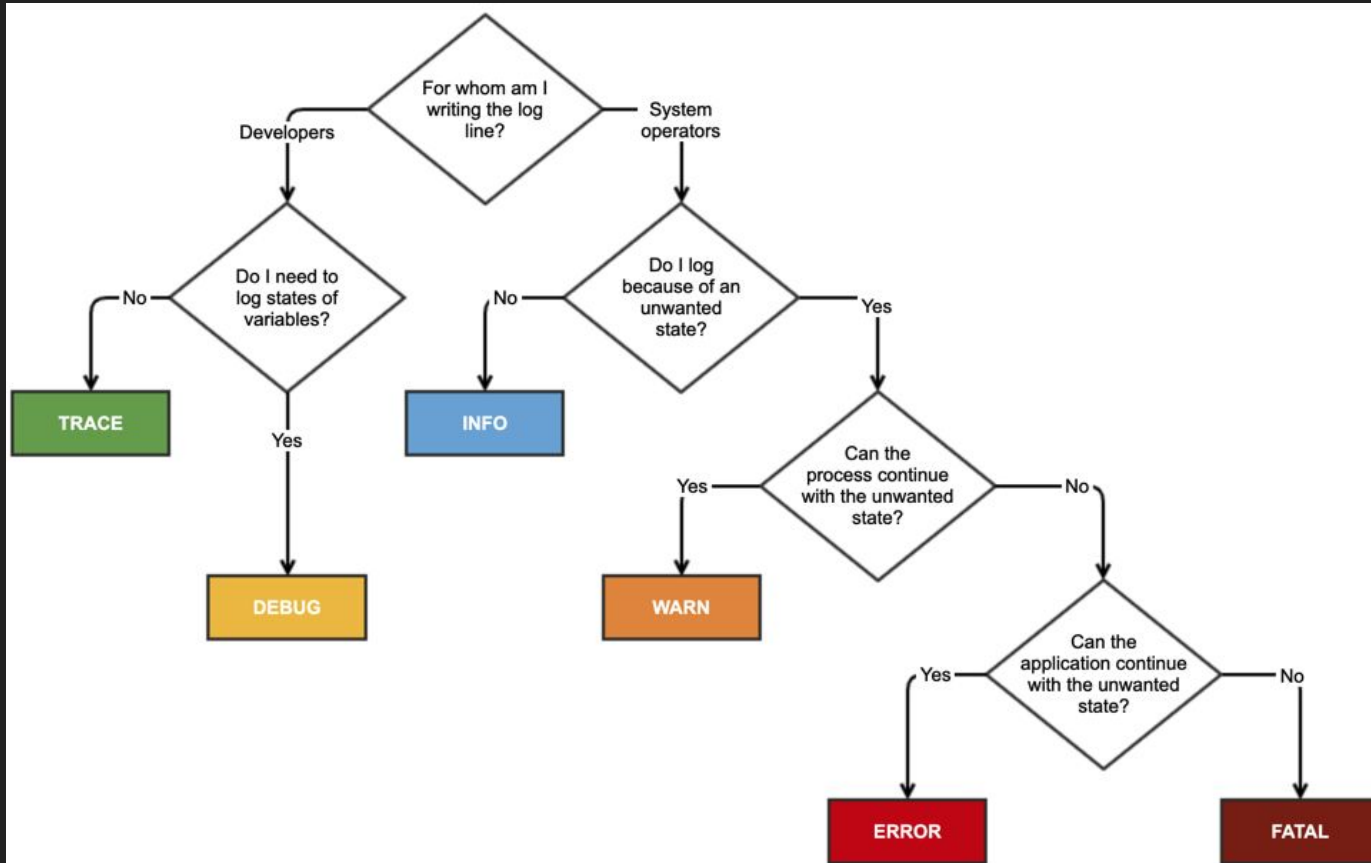
daily + size 500M => rotate on
whichever limit hits first

compress + delaycompress =>
keep the newest file plain for fast
tailing



Tip: Always create a POC for log rotate and data retention

Log Levels Example



Example #2 The TRACE Level Trap: Left On, It Costs You Twice

COST #1 — DISK & PERFORMANCE

TRACE can log every function call or query.
It fills disks fast and slows down the very
app writing it.

COST #2 — DATA EXPOSURE

TRACE routinely captures full SQL
statements, request bodies, headers —
even credentials and tokens.

```
$ grep -i level /etc/myapp/logging.conf
```

```
level = TRACE
```

```
$ tail -2 /var/log/myapp/app.log
```

```
TRACE: query: SELECT * FROM users WHERE email='bob@co.com' AND pass_hash='...'
```

```
TRACE: Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

Fix: keep default to INFO/WARN in production; TRACE only for a short, scoped debug window and save the log snippet securely.

Tip #4
Log
Aggregation
Tool

Intro to Log Aggregation Tool

- 1 The next step after Log Management
- 2 Provides Centralized Visibility 
- 3 Collect, Store & Query

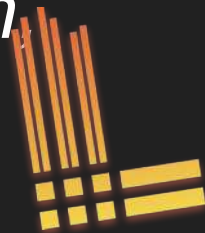
Grafana Loki

27.3K **430K**

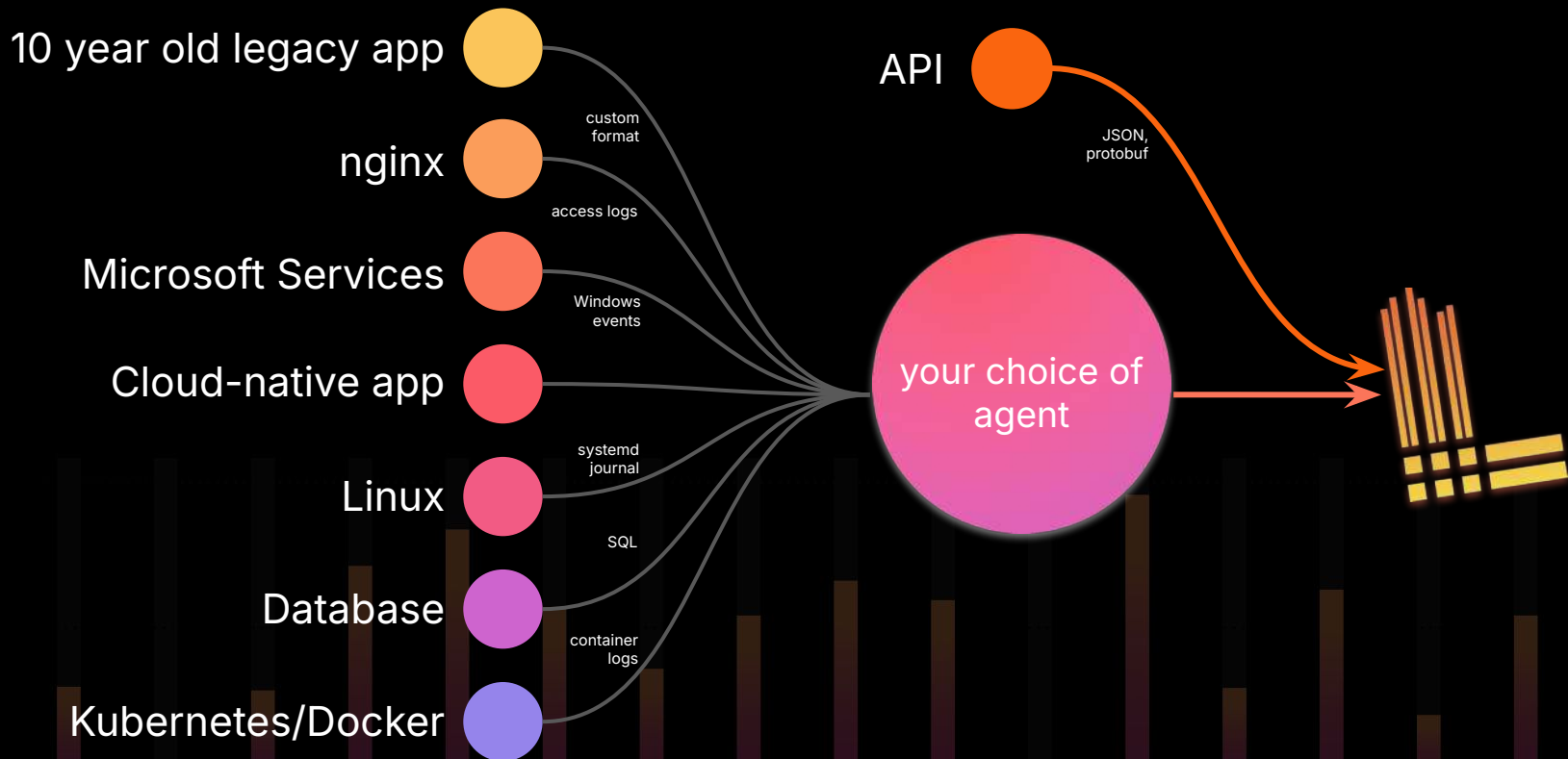
GitHub stars

Deployments

- 1 An Open Source Log aggregation tool
- 2 Provides Centralized Visibility 
- 3 Supports features such as *Alerting, Indexing/Caching, Data Visualization Annotations etc.*



Loki accepts it all



Install Loki



Deploy Agent,
Collect logs



Deploy Grafana



Query log data



Logfmt
JSON
Syslog
OTLP
Docker
Kubernetes
Kafka
Cloud
...

Compactor
Distributor
Query Frontend
Ingestor
Querier
Ruler

Drilldown Logs
Explorer
Dashboards
Alerting
...





Promtail



fluentbit



Grafana Alloy



logstash



docker



fluentd



LOGBack™
The Generic, Reliable
Fast & Flexible
Logging Framework



.NET Core



LOG4J





Promtail



Grafana Alloy



logstash



LOGBack™
*The Generic, Reliable
Fast & Flexible
Logging Framework*



How Grafana Loki does indexing

2019-12-11T10:01:02.123456789Z {app="nginx", env="dev"}

GET /about 1034 Debug "page not found"

Timestamp

with nanosecond precision

Labels/Selectors

key-value pairs

Content log line

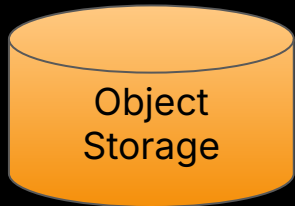
Indexed in TSDB format

Unindexed

10TB

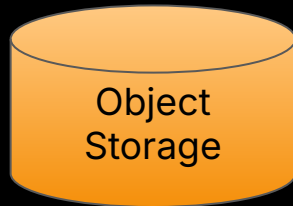
Log Data

=



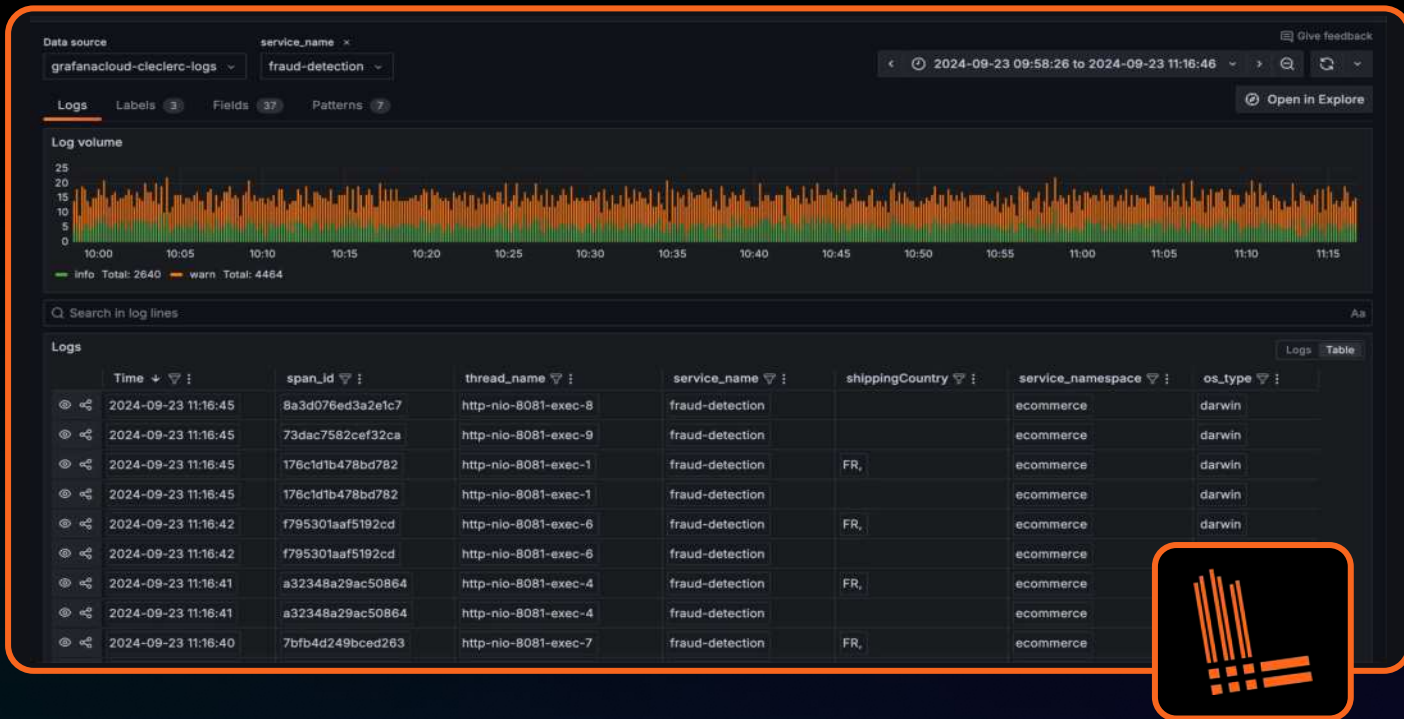
50MB+

Index



~3TB

Compressed log data



Drilldown Logs

General Availability

OSS

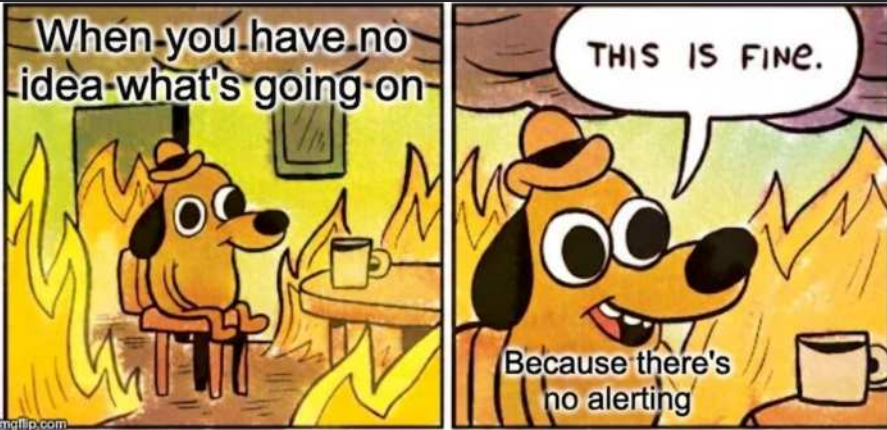
Cloud



Demo!

Tip #5
From Logs to
Alerts

Teams need to get Alerting



Alerts make it easy to **detect**, **respond**, and **learn** from incidents with minimal toil 

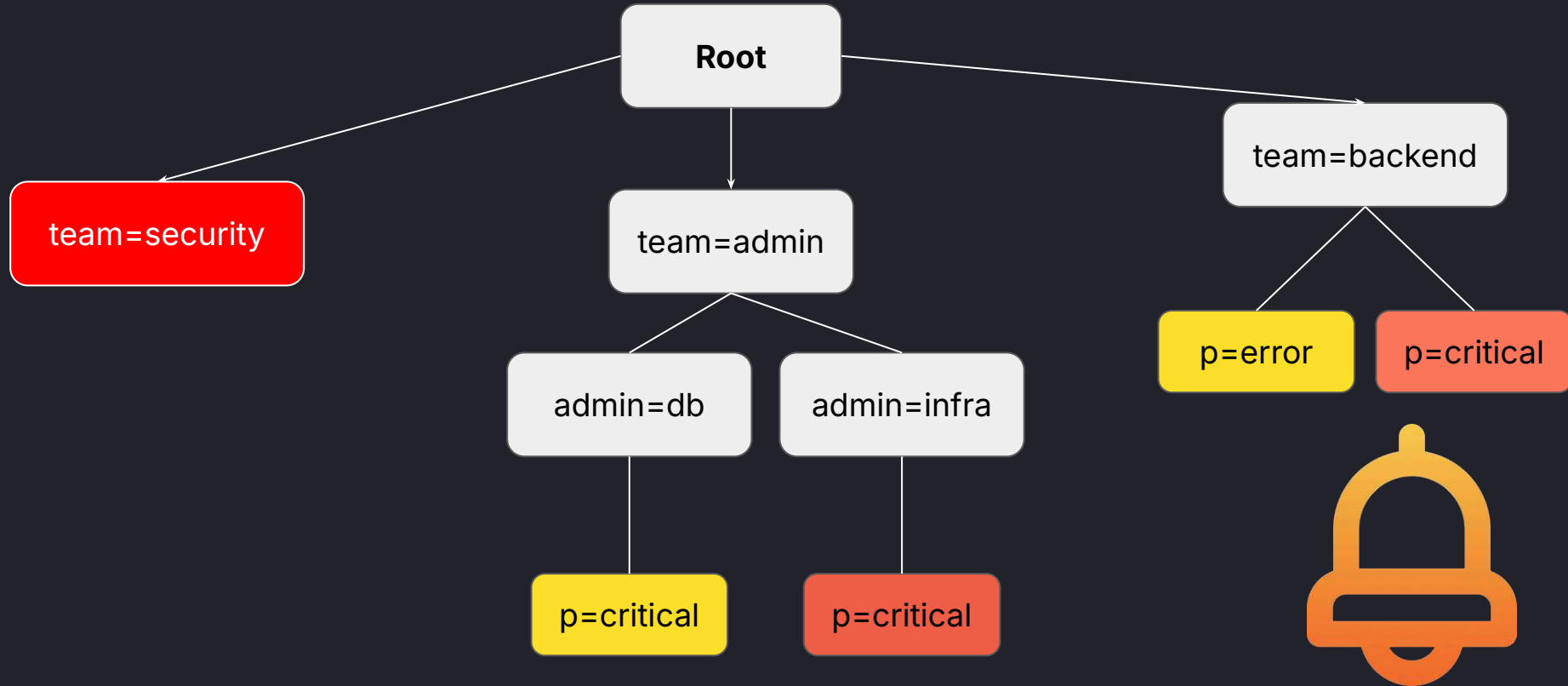


Alerting with Grafana Loki

- 1 Send alerts Easily 
- 2 Support various contact points 
- 3 Create rules, templates, policies, routes
runbooks etc.



Routing - Notification Policies





Demo!

Datasource: LokiNGINXLogs Label Name: agent Label Value: All x Job: All x Instance: All x + Get started fast with Grafana Cloud free tier

Last 6 hours Refresh

KPI's

Total requests Last 24 hours

973 k

Realtime visitors Last 5 minutes

1

Requests per status code

HTTP Status:
200
230 kHTTP Status: 307
2.27 kHTTP Status: 308
3.39 kHTTP Status: 400
3HTTP Status: 404
335

NGINX logs in bytes

295 MB

Total Bytes Sent

2.14 GB

Dashboard created by Ward Bekker (hope you like it!)

Follow me on Twitter & LinkedIn for Grafana Loki updates

Check out my Grafana Loki video's on Youtube

NGINX log lines

223 k

% of 5xx requests

No data

% of requests by Googlebot

0.188%

HTTP status codes over time



Top Countries

BD

BR

CN

EC

FR

GB

RU

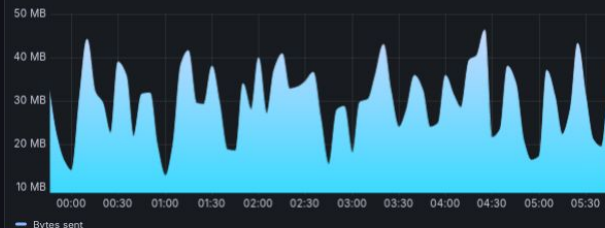
SG

US

Recent requests

```
GET /a/6503121535/alternative-to-next-ink-ai-tattoo-design.html?page=31 with HTTP status: 200
GET /a/1494460811/alternative-to-idle-space-miner-cash-empire.html with HTTP status: 200
GET /?q=justphoto with HTTP status: 200
GET /a/1092694950/alternative-to-kbcs-public-radio-app.html?page=550 with HTTP status: 200
GET /a/1547107922/alternative-to-soda-bottle-flip.html?page=29 with HTTP status: 200
GET /a/6711361052/alternative-to-us-sales-tax-calculator.html?page=74 with HTTP status: 200
GET /a/1336163583/alternative-to-next-up-performance.html?page=73 with HTTP status: 200
GET /a/1477805164/alternative-to-hiv.html with HTTP status: 307
GET /?q=jumpy%20head with HTTP status: 200
GET /a/1437432687/alternative-to-zion-public-school-ok.html?page=537 with HTTP status: 200
GET /a/1026242937/alternative-to-supercines.html?page=281 with HTTP status: 200
GET /?q=ivy%20ties with HTTP status: 200
```

Bytes Sent



95th percentile of Request Time

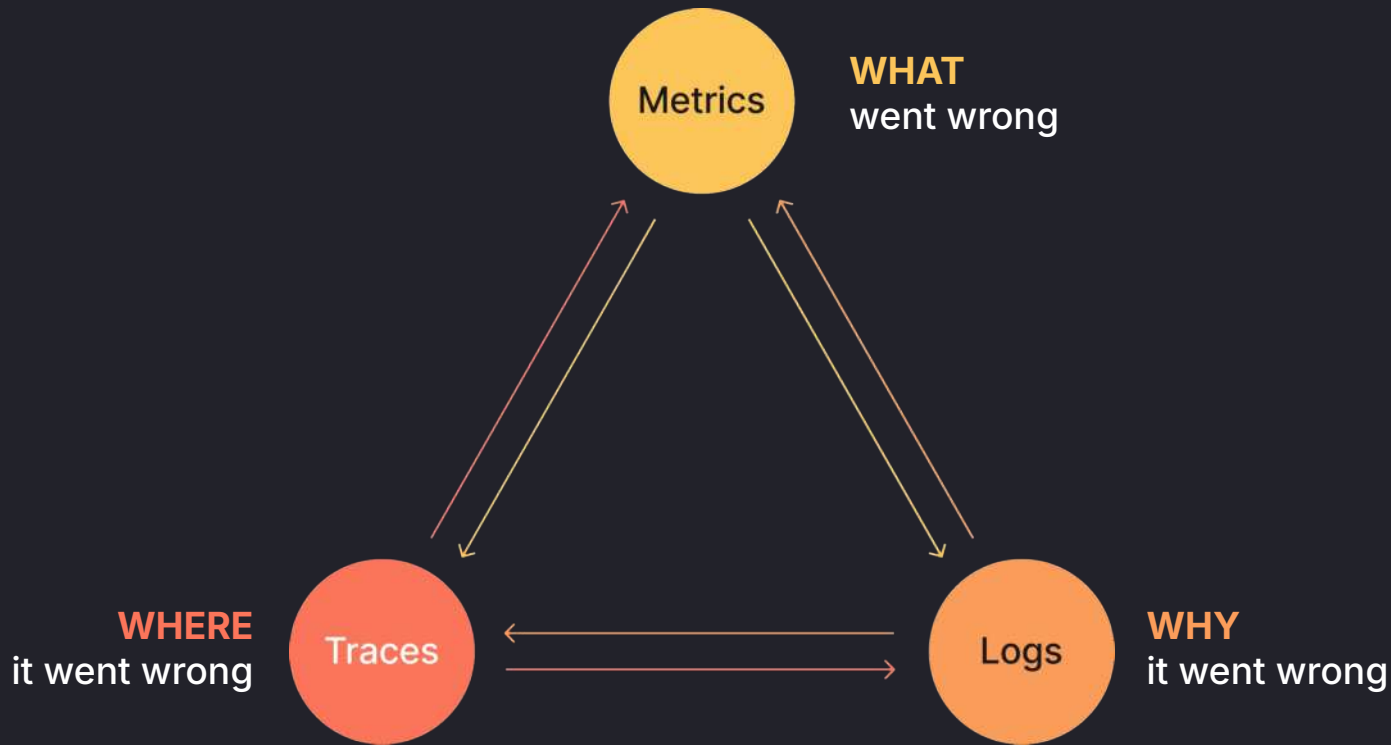


Loki based NGINX
Dashboard publicly
available on Grafana Play



Additional Tip:
**Vendor Neutral
Approach for
your
Infrastructure**

Pillars of Observability

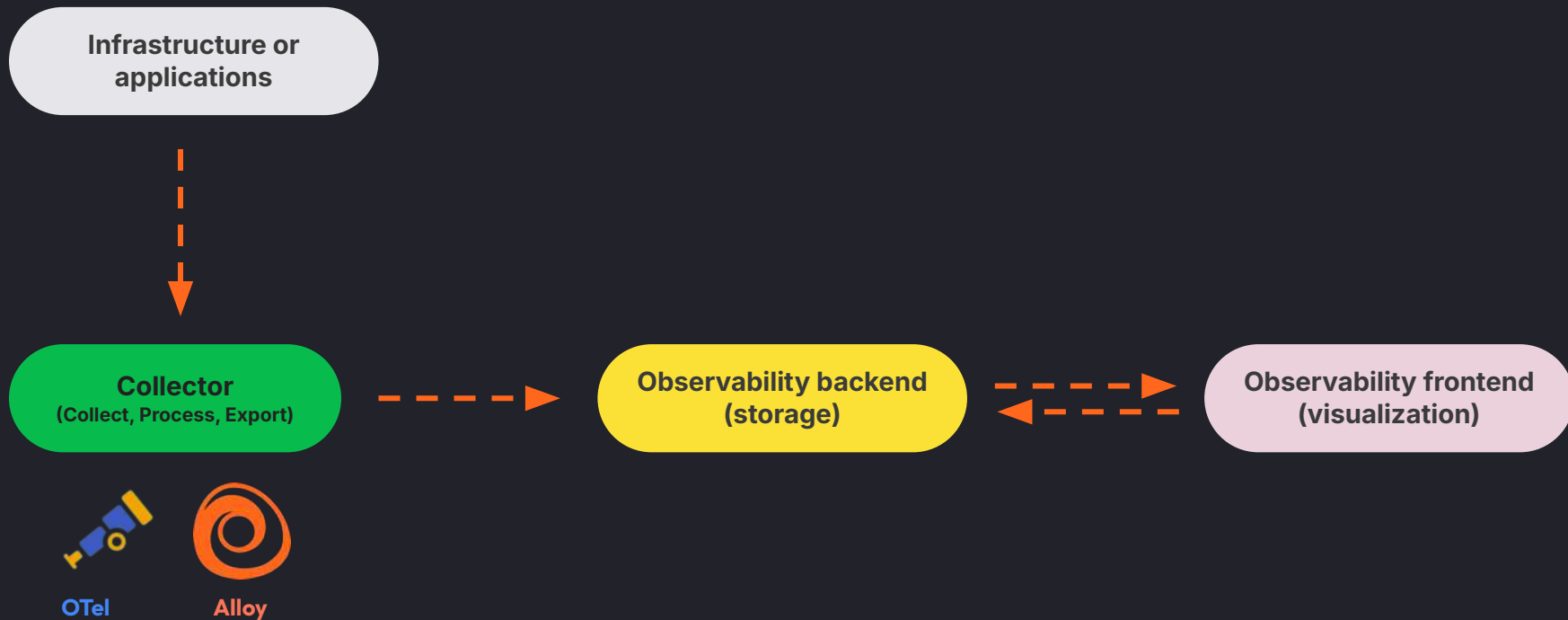


Vendor Neutral Approach

- 1 Public or Private Infrastructure Problems
- 2 Your solution does not work for others
- 3 Vendor-Neutral and Cross Platform

e.g. **OTel** ( *CNCF Graduated Project*)

Overview

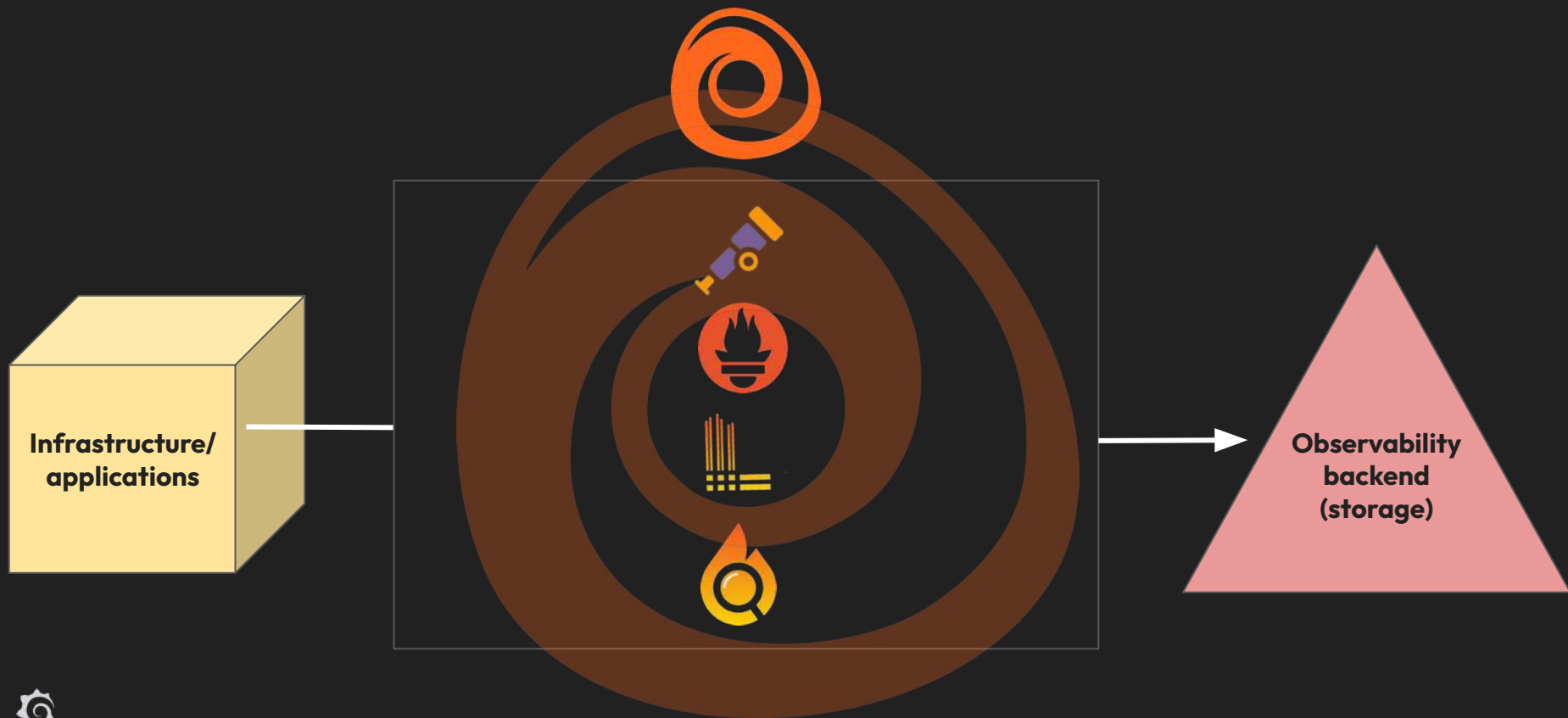


Grafana Alloy

- 1 Core component in Grafana stack
- 2 **100%** OTel compatible + Community
- 3 Provides more features than OTel

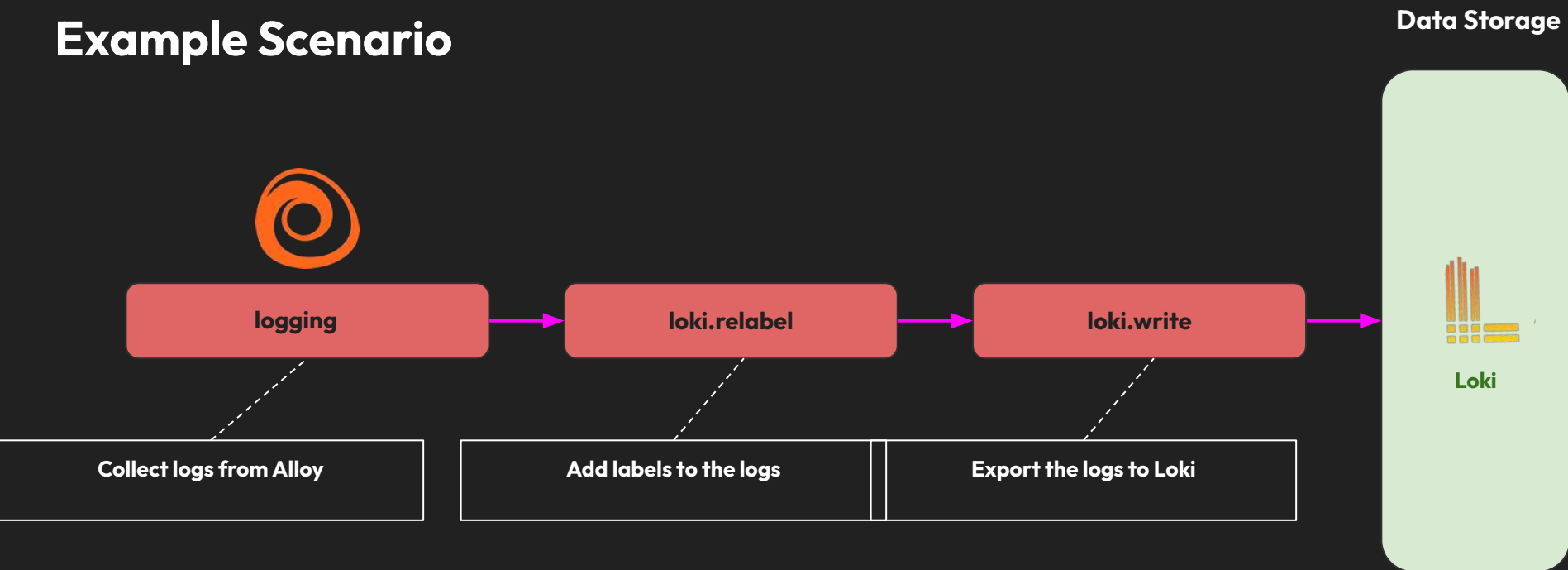


Using Grafana Alloy (OTel)



Build a visible pipeline for infrastructure logs

Example Scenario



Hot Reload!!

Reloading the config

To reload Alloy's config, hit the following endpoint in a browser or with a tool like `curl` :

```
curl -X POST http://localhost:12347/-/reload
```



If the config is valid, you should see a response like the following:

```
config reloaded
```





Demo!

Final Takeaways

Three circles are arranged horizontally. The first circle on the left has a blue border and contains the text 'Practice the core'. The middle circle has a green border and contains the text 'Centralized Visibility is IMPORTANT!!'. The third circle on the right has a magenta border and contains the text 'Document every step for RCA's & Postmortem'.

Practice the
core

Centralized
Visibility is
IMPORTANT!!

Document
every step for
RCA's &
Postmortem



Thanks for participating!

Contact me:



usman.ahmad@grafana.com



linkedin.com/in/usmanlinux/

